

Cross-Sectional Momentum Under Implementation Stress

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Code and data pipeline: <https://github.com/scalex-research/scalex-momentum-research>

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Abstract

We ask whether a classical cross-sectional momentum signal stays statistically and economically meaningful when reasonable implementation choices change. The signal is the trailing 12-month return skipping the most recent month. The portfolio is long the top quintile and short the bottom quintile, rebalanced monthly. The assets are the 49 US industry portfolios of the Kenneth R. French Data Library, with holding months from 1927-07 to 2026-08 (1,190 months).

We fixed the primary specification, 20 one-at-a-time variations, the tests and a decision rule in writing before computing any result.

Gross of costs, the primary long-short portfolio earned **7.95% per year** (Newey-West $t = 4.59$). Net of an assumed 25 bps per unit of turnover, it earned **4.96%** ($t = 2.83$, $p = 0.005$). After a Holm correction across the 21 pre-defined specifications, the net result is **not significant at 5%** (adjusted $p = 0.052$).

The gross effect is significant in 15 of 16 non-cost variations. The net effect is fragile. It disappears for every 3- and 6-month lookback and for the coarsest (12-industry) universe. The break-even cost is 67 bps. The effect is absent after three-year market declines, where the worst losses also cluster.

By the criterion we stated in advance, the result is **partially robust**. All results are for industry portfolios, not individual stocks, and rest on a stylised cost model.

1. Research question

Cross-sectional momentum is the tendency of assets with high recent returns to keep outperforming assets with low recent returns over the following months. It is one of the most studied regularities in empirical finance (Jegadeesh & Titman, 1993; Carhart, 1997; Asness, Moskowitz & Pedersen, 2013). Among industries specifically, it was documented by Moskowitz & Grinblatt (1999).

We do not ask whether momentum "exists"; the literature has asked that many times on overlapping data. We ask a narrower and, for practitioners, more useful question:

Does a simple cross-sectional momentum signal remain statistically and economically meaningful when we vary universe construction, ranking methodology, portfolio formation, transaction costs, holding period and other implementation choices?

A published backtest is one specification picked from many plausible ones. If the conclusion survives only under a narrow set of choices, a reader should hold it more loosely than its headline t-statistic suggests. This project measures that fragility directly.

2. Hypotheses

The plan in [docs/preregistration.md](#) was written before any result was computed. It is reproduced here in short form.

Let r_t be the monthly return of the primary long-short portfolio.

- **H1 (statistical):** $E[r_t, \text{gross}] \neq 0$, with a positive sign expected. **H0:** $E[r_t, \text{gross}] = 0$.
- **H1-net (economic):** the same, net of the cost model.
- **Test:** two-sided t-test of the mean with Newey-West HAC standard errors, lag $L = \text{floor}(4(T/100)^{(2/9)}) = 6$, $\alpha = 5\%$.
- **Robustness criterion (fixed in advance).** The effect is called *robust* only if all three conditions hold:
 1. The primary net result is positive and significant.
 2. At least 75% of the 16 non-cost variations leave the conclusion unchanged. A variation "changes the conclusion" if the sign of the mean net return, or its 5% significance, differs from the primary.
 3. The mean net return is positive in each of three equal-length subperiods.

What pre-registration does and does not buy here. The plan was fixed before computing results, but not in ignorance of the literature. Industry momentum is known to be positive over long US samples, and parameter choices such as "12-1" and quintiles are conventions, partly because they worked in earlier studies on overlapping data. This study is therefore **not out-of-sample evidence** for momentum. Its contribution is a fixed, transparent protocol and a measurement of sensitivity.

3. Data

Item	Detail
Source	Kenneth R. French Data Library: 49, 30, 17 and 12 Industry Portfolios (monthly); Fama/French 3 Factors (for the T-bill rate and market return); Momentum Factor (exploratory comparison only)
URL	https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html
Vintage / access	Files built from the CRSP 202608 database; downloaded 2026-10-01. SHA-256 hashes are in <code>results/run_metadata.json</code>
Coverage	July 1926 – August 2026, monthly; returns in percent, converted to decimals
Construction	Each NYSE, AMEX and NASDAQ stock is assigned to an industry by its four-digit SIC code; industry returns are value-weighted (primary) or equal-weighted (variation R11)
Missing data	Coded –99.99/–999 and converted to missing; never filled. Nine of the 49 industries start after July 1926 (five in July 1963). <code>Rubbr</code> has a 12-month gap in 1943–44. No eligible position had a missing holding-month return (0 cases)
Licensing	The data are © Fama and French and are not redistributed in the repository. Users download them with <code>scripts/download_data.py</code>

Survivorship. The industry portfolios include all CRSP stocks in each month, including stocks that later delist, so the cross-section is not filtered on survival. French's documentation does not state in detail how delisting returns enter the industry returns, so we cannot rule out a small residual bias.

A material data feature. Before the 1960s, several of the 49 "industries" contain only one or two firms. In August 1932 the one-firm *Paper* industry returned +300% in a single month. The primary universe therefore mixes genuine industry baskets with single stocks in the early sample. We did not change the pre-registered universe. Instead we report an **exploratory** check that excludes industries with fewer than five firms at formation. Its net mean return is 4.97% per year ($t = 3.01$), close to the primary.

Full details: <docs/data.md>.

4. Methodology

4.1 Timing (no look-ahead)

The row for month t holds the return earned during month t , known at the close of month t . At the close of each formation month t :

1. **Observe** returns through month t .
2. **Compute the signal:** the cumulative return over months $t-11 \dots t-1$, i.e. $s_{\{i,t\}} = \prod_{k=1}^{11} (1 + r_{\{i,t-k\}}) - 1$. Month t itself is skipped.
3. **Check eligibility:** an asset is eligible if it has a valid return in every month $t-11 \dots t$. The same 12-month rule applies to every lookback, so changing the signal never changes the universe.
4. **Rank** eligible assets on $s_{\{i,t\}}$. Ties are broken by fixed column order. With N eligible assets, $\text{bucket} = \lfloor \text{rank} \times 5 / N \rfloor$.

5. **Form portfolios:** equal weights within buckets, long the top quintile (+1 in total) and short the bottom quintile (−1 in total).
6. **Measure** the portfolio's return over month $t+1$.

The code keeps formation-month and holding-month indices separate, and the test suite enforces the order. Rewriting every return after month T leaves every signal, weight and regime label dated T or earlier unchanged. On pure noise the pipeline finds no effect. A deliberately injected one-month leak drives the HAC t-statistic above 20, which shows that the test would catch one.

4.2 Portfolio return, turnover and costs

With weights $w_{\{i,t\}}$ formed at t , the gross return is $R_{\{t+1\}} = \sum_i w_{\{i,t\}} r_{\{i,t+1\}}$.

Turnover traded at the close of t is $TO_t = \sum_i |w_{\{i,t\}} - \tilde{w}_{\{i,t\}}|$. The drifted weights $\tilde{w}_{\{i,t\}} = w_{\{i,t-1\}}(1 + r_{\{i,t\}}) / (1 + R_t)$ are last month's weights after one month of returns. TO counts purchases and sales in both legs, so replacing both legs completely gives $TO = 4$.

The net return is $R_{\{t+1\}} - c \times TO_t$, with $c = 25$ bps per unit traded in the primary specification. The primary portfolio holds about 9.6 industries per leg and trades an average TO of 1.00 per month. That costs 2.99% per year at 25 bps.

The cost model is deliberately simple. It is a fixed charge per dollar traded, not a market-impact model (Section 11).

For **multi-month holding periods** ($K = 3, 6$), we use the overlapping-cohort construction of Jegadeesh & Titman (1993). The book holds the average of the last K formation-date portfolios, and one- K th of it is re-formed each month. This yields a single non-overlapping monthly return series. We never compute statistics on overlapping K -month returns.

4.3 Statistics

Statistic	Definition
Annualised mean	$12 \times \text{mean monthly return (arithmetic)}$
CAGR	$(\prod (1+r))^{\{12/T\}} - 1$
Annualised volatility	$\sqrt{12} \times \text{monthly standard deviation}$
Sharpe ratio	annualised mean excess return / annualised volatility. A long-short return is already an excess return; long-only portfolios subtract the 1-month T-bill rate
Maximum drawdown	worst peak-to-trough decline in cumulative wealth
Hit rate	share of months with a positive return
Downside deviation / Sortino	$\sqrt{12} \times \text{root-mean-square of negative excess returns; annualised mean excess return over it}$
Turnover	TO_t as defined above, averaged over months

Why Newey-West? Monthly long-short returns are heteroskedastic: volatility in the 1930s was several times its post-war level. They are also mildly autocorrelated. The Ljung-Box test on 12 lags rejects zero autocorrelation for the primary gross series ($p = 0.006$), and overlapping-cohort variations are autocorrelated by construction. HAC standard errors remain valid under both conditions. For K -month holding the lag is at least K .

Sharpe-ratio confidence intervals come from a stationary block bootstrap (Politis & Romano, 1994) with expected block length 12, 2,000 resamples and seed 20261001.

The $\sqrt{12}$ scaling of the Sharpe ratio assumes no serial correlation. Lo's (2002) adjustment changes the primary net Sharpe ratio by a factor of 1.02, which is immaterial.

5. Primary results (confirmatory)

Portfolio	Ann. mean	Ann. vol	Sharpe [95% CI]	HAC t	Max DD	Turnover/mo
Long-short, gross	7.95%	17.8%	0.45 [0.23, 0.71]	4.59	-87.3%	1.00
Long-short, net of 25 bps	4.96%	17.8%	0.28 [0.08, 0.51]	2.83	-91.0%	1.00
Winners (Q5), excess of T-bill	13.28%	20.6%	0.64	6.13	-77.0%	0.47
Losers (Q1), excess of T-bill	5.33%	26.3%	0.20	1.99	-91.5%	0.47
Market (CRSP VW), excess of T-bill	8.27%	18.4%	0.45	4.33	-83.7%	–

The full table, with CAGR, hit rate, skewness, kurtosis, downside deviation and Sortino ratio, is in [results/tables/primary_performance.md](#).

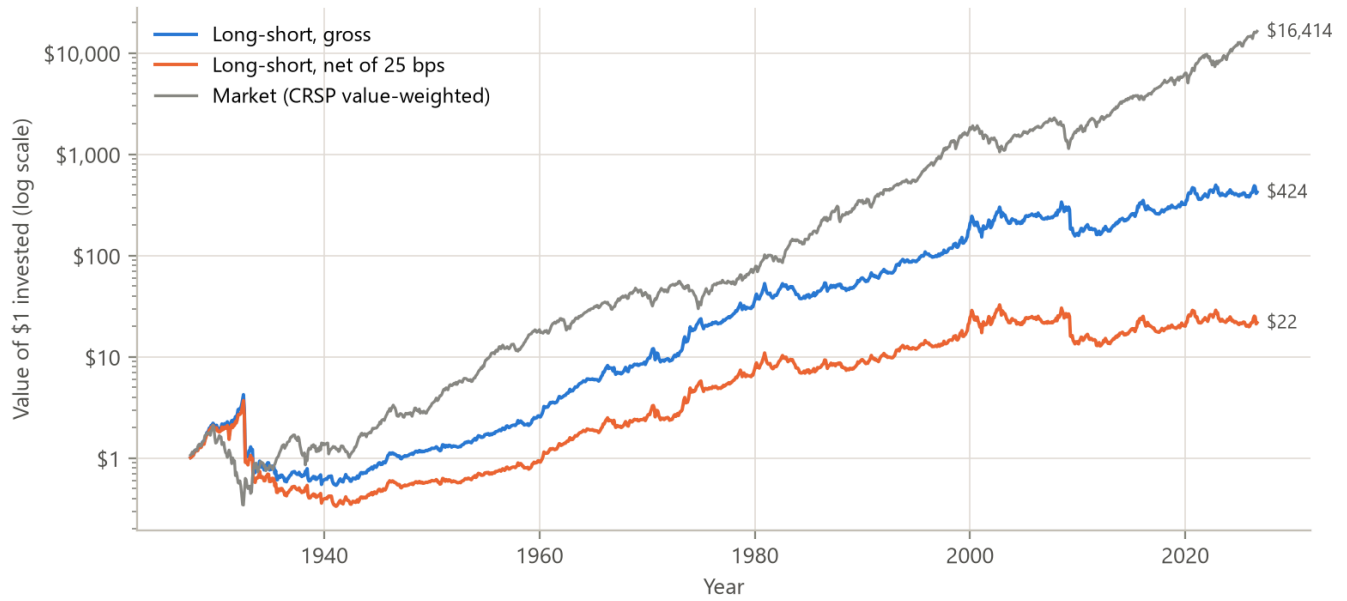
Gross. H_0 is rejected. The mean gross long-short return is 7.95% per year, with a 95% HAC confidence interval of [4.55%, 11.34%]. Average excess returns rise monotonically from the loser to the winner quintile (Figure 10).

Net. H_0 is also rejected at the single-test level: $p = 0.005$, interval [1.52%, 8.40%]. The lower bound of that interval is economically small.

Risk profile. The net long-short portfolio had a Sharpe ratio of 0.28, below the 0.45 of the market itself over the same period. Its return distribution is strongly left-skewed (skewness -2.77, excess kurtosis 28.1). The worst month was -64.4%, and the maximum drawdown of -91.0% occurred in the early 1930s. The mean conceals crash risk that a Sharpe ratio does not capture.

Cumulative growth of \$1

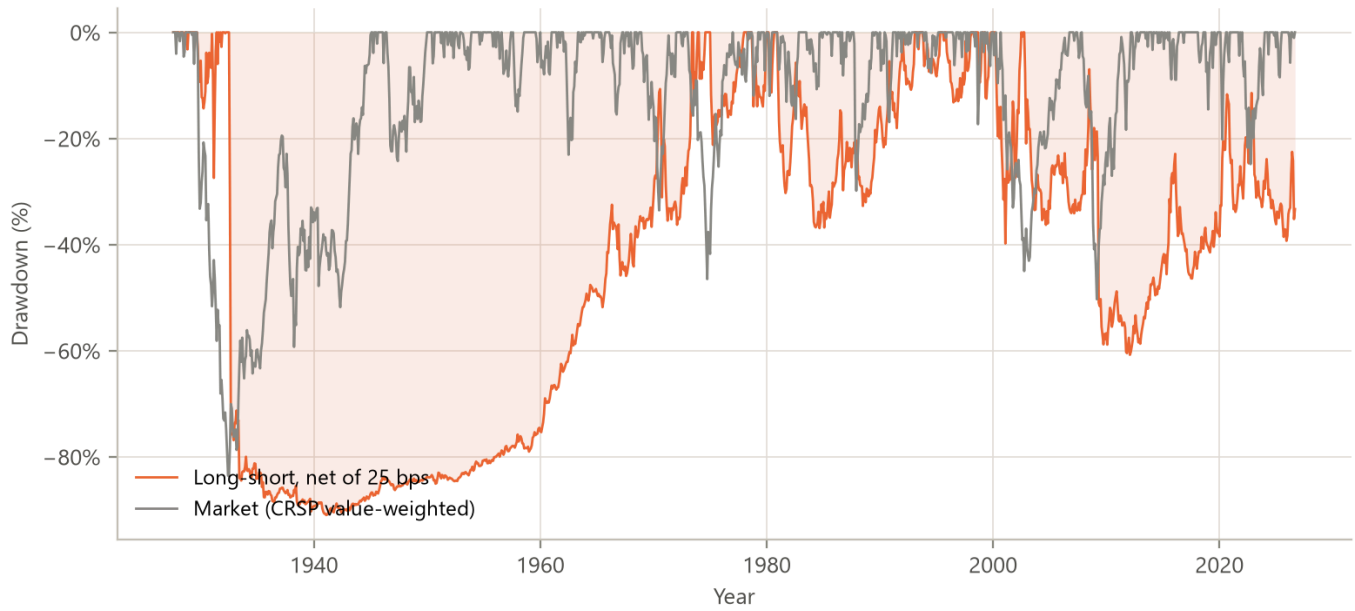
Primary long-short (top minus bottom quintile) vs. the market, log scale



Source: Kenneth R. French Data Library (CRSP-based); ScaleX calculations. Sample: holding months 1927-07 to 2026-08. Long-short returns are zero-investment returns on \$1 per leg; market is total return. Simulated, before taxes, financing spreads and borrow fees.

Drawdowns

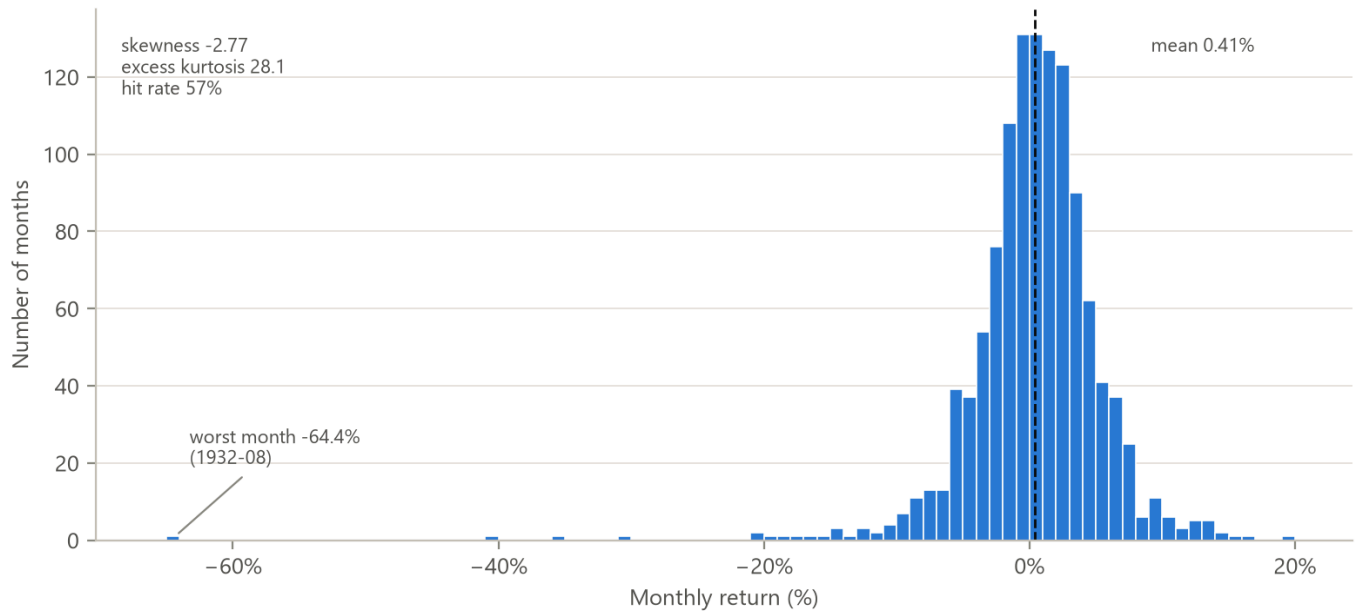
Decline from previous peak of cumulative wealth



Source: Kenneth R. French Data Library (CRSP-based); ScaleX calculations. Sample: holding months 1927-07 to 2026-08.

Distribution of monthly returns

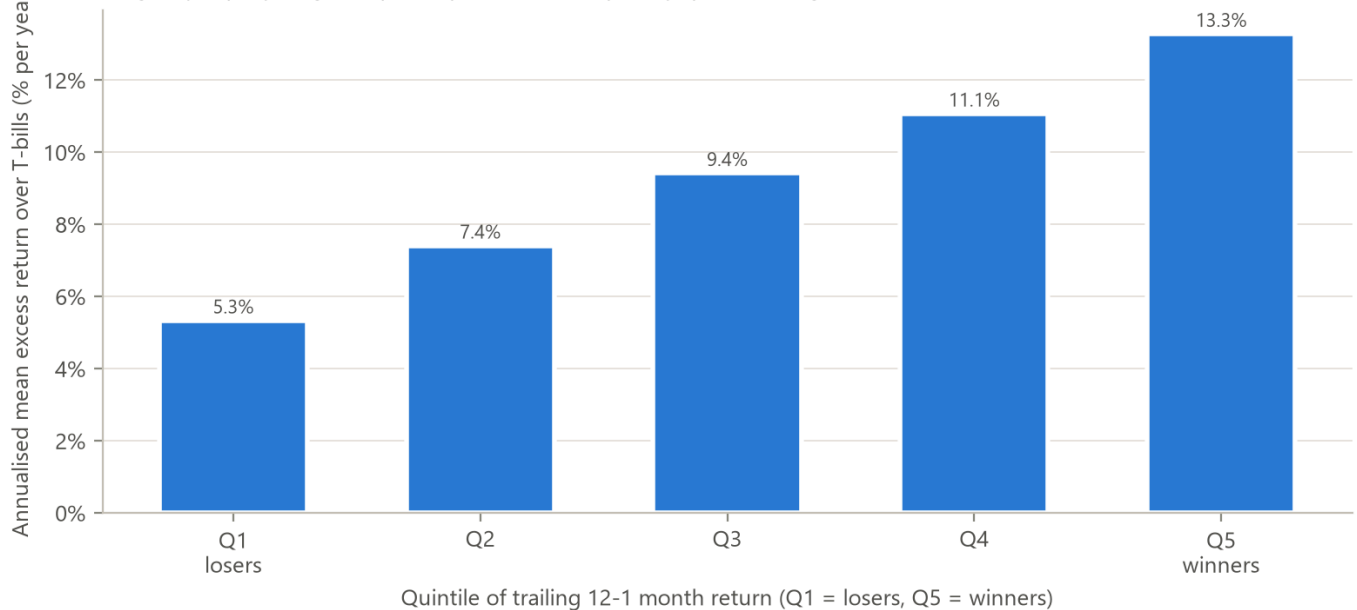
Primary long-short portfolio, net of 25 bps cost per unit turnover



Source: Kenneth R. French Data Library (CRSP-based); ScaleX calculations. Sample: holding months 1927-07 to 2026-08.

Average excess return by momentum quintile

Long-only, equally weighted quintile portfolios of the primary specification, gross



Source: Kenneth R. French Data Library (CRSP-based); ScaleX calculations. Sample: holding months 1927-07 to 2026-08.

6. Transaction costs

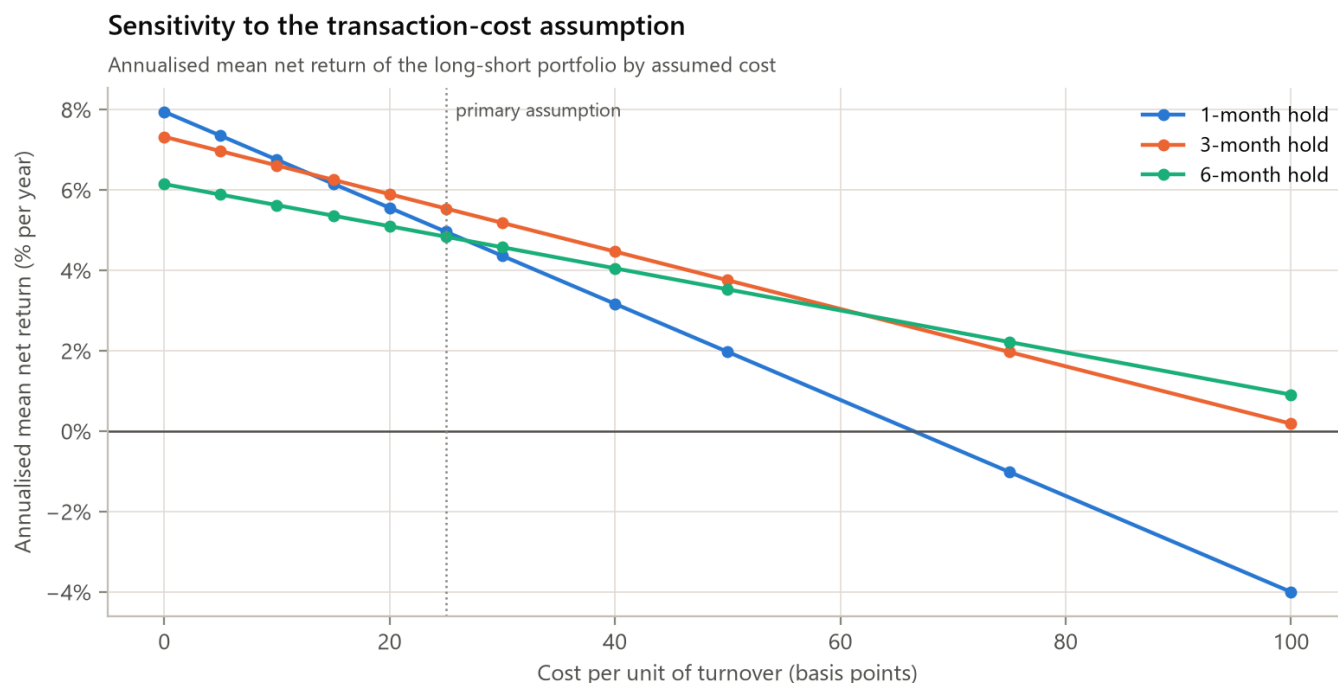
Costs reduce the mean return by 2.99% per year at the primary assumption.

Break-even cost. The mean net return reaches zero at 67 bps per unit of turnover.

Cost (bps per unit traded)	1-month hold	3-month hold	6-month hold
0	7.95% (t 4.59)	7.32% (t 4.49)	6.15% (t 3.90)
10	6.75% (t 3.88)	6.61% (t 4.04)	5.62% (t 3.55)
25 (primary)	4.96% (t 2.83)	5.54% (t 3.36)	4.84% (t 3.04)
50	1.97% (t 1.11)	3.75% (t 2.26)	3.53% (t 2.20)
100	-4.00% (t -2.19)	0.18% (t 0.11)	0.91% (t 0.56)

Longer holding periods trade less: average turnover is 0.59 for 3 months and 0.44 for 6 months, against 1.00 for 1 month. Their gross returns are lower, but they degrade more slowly as costs rise. The cost level at which the rankings swap depends entirely on the assumed cost, which we do not observe.

Which cost assumption is right? For large, liquid US stocks traded by institutions after about 2000, published estimates of all-in costs are often in the 5–20 bps range. Costs before the 1970s were much higher; for the early twentieth century, Jones (2002) estimates one-way costs near or above 1%. A single number across 99 years is therefore too generous early in the sample and possibly too harsh late. The primary cost of 25 bps is a middle assumption; it is **not** an estimate.



Source: Kenneth R. French Data Library (CRSP-based); ScaleX calculations. Sample: holding months 1927-07 to 2026-08. Stylised proportional cost model; not an estimate of market impact.

7. Robustness

Each row changes exactly one choice. "Changes" means the pre-defined conclusion differs from the primary: the sign or the 5% significance of the net mean. Net figures use 25 bps unless the row changes the cost.

ID	Variation	Gross (t)	Net (t)	Net 95% CI	Turnover	Break-even	Changes?
P0	Primary	7.95% (4.59)	4.96% (2.83)	[1.5%, 8.4%]	1.00	67 bps	–
R01	Signal 3-0	3.76% (2.47)	-1.76% (-1.13)	[-4.8%, 1.3%]	1.84	17 bps	yes
R02	Signal 3-1	1.14% (0.70)	-5.56% (-3.38)	[-8.8%, -2.3%]	2.23	4 bps	yes
R03	Signal 6-0	4.86% (2.68)	0.87% (0.47)	[-2.8%, 4.5%]	1.33	30 bps	yes
R04	Signal 6-1	4.47% (2.53)	0.11% (0.06)	[-3.4%, 3.6%]	1.45	26 bps	yes
R05	Signal 12-0	8.68% (4.78)	5.81% (3.15)	[2.2%, 9.4%]	0.96	76 bps	no
R06	3-month holding	7.32% (4.49)	5.54% (3.36)	[2.3%, 8.8%]	0.59	103 bps	no
R07	6-month holding	6.15% (3.90)	4.84% (3.04)	[1.7%, 8.0%]	0.44	117 bps	no
R08	Terciles	6.56% (5.12)	4.20% (3.24)	[1.7%, 6.7%]	0.79	69 bps	no
R09	Deciles	11.34% (4.87)	7.55% (3.18)	[2.9%, 12.2%]	1.26	75 bps	no
R10	Inverse-volatility weights	7.34% (4.52)	4.08% (2.48)	[0.9%, 7.3%]	1.09	56 bps	no
R11	49 ind., equal-weighted returns	10.92% (6.59)	8.11% (4.85)	[4.8%, 11.4%]	0.94	97 bps	no
R12	30 industries	7.65% (4.59)	4.72% (2.80)	[1.4%, 8.0%]	0.98	65 bps	no
R13	17 industries	6.29% (4.01)	3.37% (2.12)	[0.3%, 6.5%]	0.98	54 bps	no
R14	12 industries	5.40% (3.67)	2.31% (1.54)	[-0.6%, 5.2%]	1.03	44 bps	yes
R15	Cost 0 bps	7.95% (4.59)	7.95% (4.59)	[4.6%, 11.3%]	1.00	67 bps	no
R16	Cost 10 bps	7.95% (4.59)	6.75% (3.88)	[3.3%, 10.2%]	1.00	67 bps	no
R17	Cost 50 bps	7.95% (4.59)	1.97% (1.11)	[-1.5%, 5.5%]	1.00	67 bps	yes
R18	Cost 100 bps	7.95% (4.59)	-4.00% (-2.19)	[-7.6%, -0.4%]	1.00	67 bps	yes
R19	Winsorised 1%/99%	9.11% (6.24)	6.13% (4.17)	[3.2%, 9.0%]	1.00	67 bps	no
R20	Post-1963 formation	8.32% (3.99)	5.38% (2.56)	[1.3%, 9.5%]	0.98	71 bps	no

Reading the table:

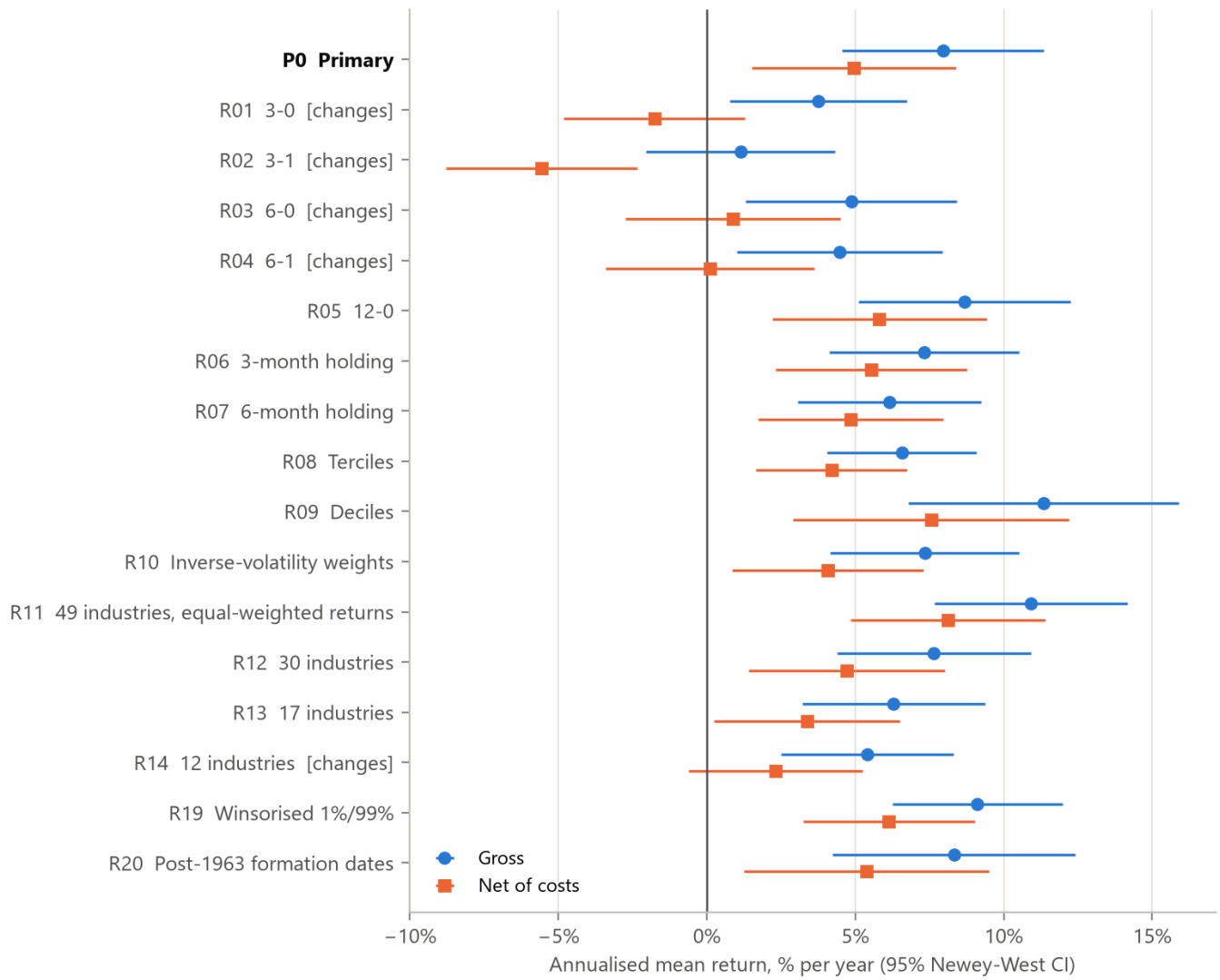
- **Gross existence is robust.** 15 of the 16 non-cost variations have a positive, significant gross mean. The exception is the 3-1 signal (R02).
- **Net significance is not.** It survives in 11 of the 16 non-cost variations. It fails for every 3- and 6-month lookback (R01–R04) and for the 12-industry universe (R14).
- **Short lookbacks are cost-dominated.** The 3-1 signal (R02) trades 2.23 per month and is *significantly negative* net of costs (-5.56%, $t = -3.38$). The highest break-even cost among the shorter lookbacks is 30 bps (R03, 6-0).
- **Construction choices with stronger results.** Deciles (R09) and equal-weighted industry returns (R11) produce larger means. We report them, but we do not promote them; R11 in particular weights small stocks heavily, and those are the most expensive to trade, which our flat cost model ignores.
- **Coarser universes are weaker.** The effect shrinks from 49 to 30, 17 and 12 industries. With 12 industries, each quintile holds two or three industries.
- **Outliers work against the strategy.** Winsorising (R19) *raises* the t-statistic. The extreme months are mostly large losses, so trimming them helps.

Pre-registered verdict: partially robust. - Condition 1 (primary net positive and significant): **met**. - Condition 2: **not met**. Only 11 of 16 non-cost variations (68.8%) leave the conclusion unchanged, short of the 75%

threshold. - Condition 3 (positive net mean in all three subperiods): **met**.

Robustness: one change at a time

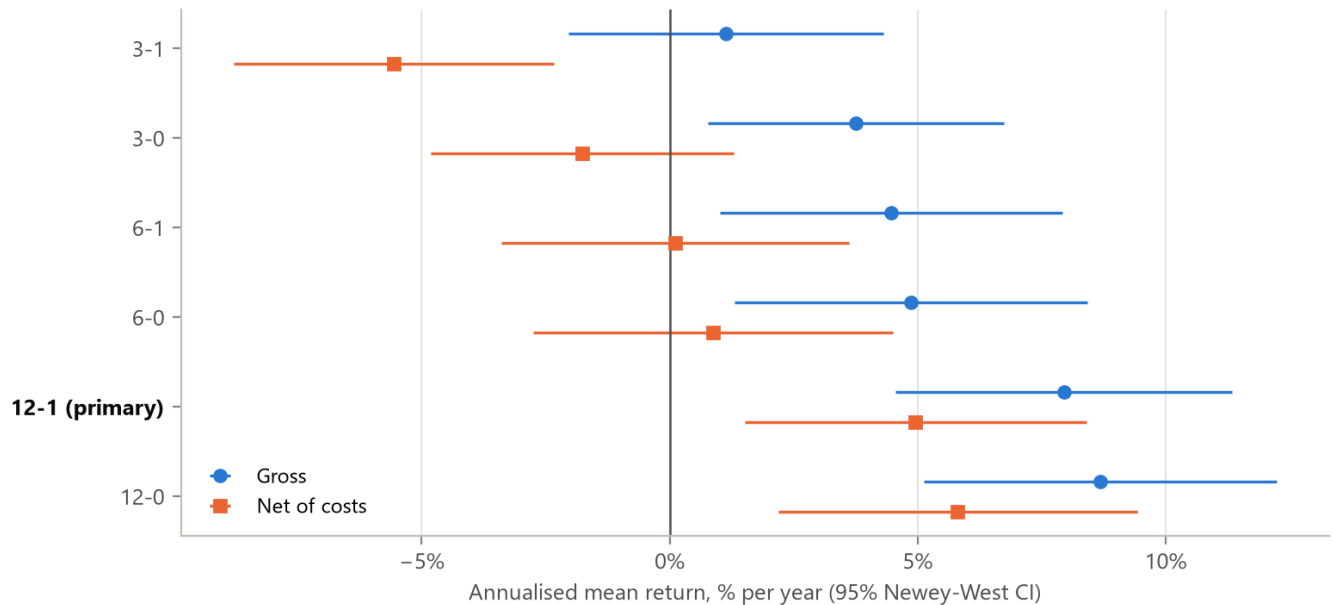
Annualised mean long-short return, gross and net of 25 bps, with 95% Newey-West intervals



Source: Kenneth R. French Data Library (CRSP-based); ScaleX calculations. Sample: holding months 1927-07 to 2026-08. [changes] = the pre-registered conclusion (sign or 5% significance of the net mean) differs from the primary.

Performance by momentum signal definition

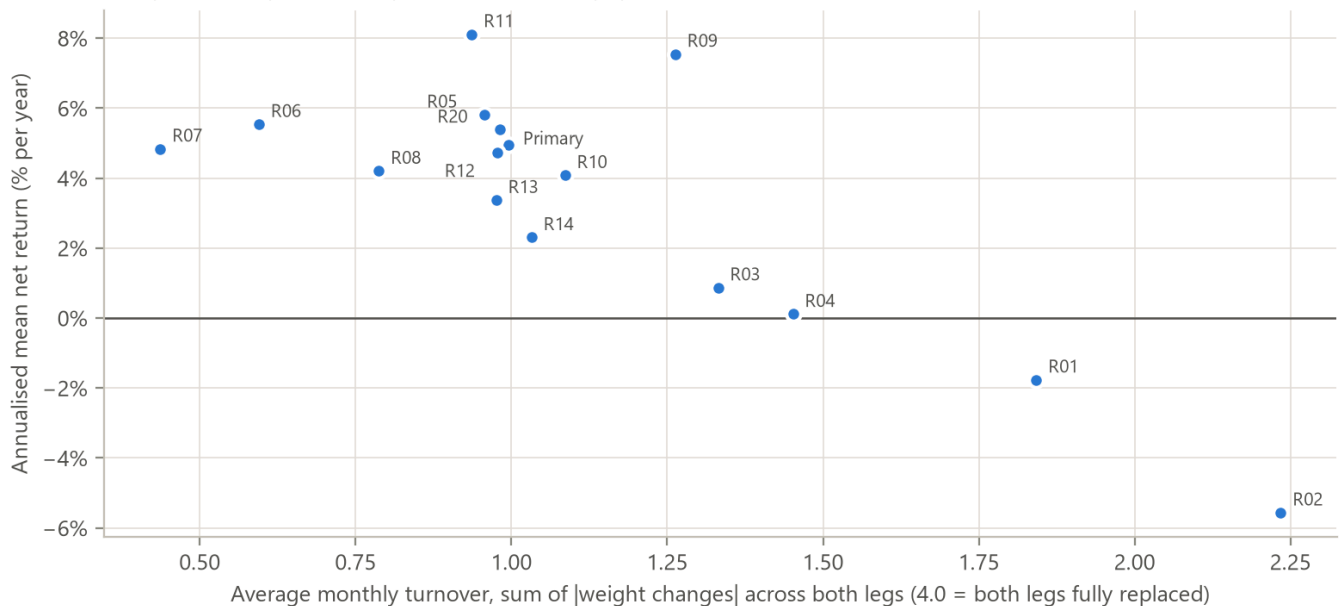
Lookback-skip in months; long-short quintiles, 1-month holding, net of 25 bps



Source: Kenneth R. French Data Library (CRSP-based); ScaleX calculations. Sample: holding months 1927-07 to 2026-08. Signal = cumulative return over months $t-L+1..t-S$, formed at the close of month t .

Turnover versus net performance

Each point is one pre-defined specification (cost 25 bps per unit turnover)



Source: Kenneth R. French Data Library (CRSP-based); ScaleX calculations. Sample: holding months 1927-07 to 2026-08. Cost variations (R15-R18) and the winsorised test (R19) share the primary's turnover and are omitted.

8. Statistical inference and multiple testing

This project runs 21 pre-defined specifications. Even on pure noise, some would cross $p < 0.05$. Applying the Holm-Bonferroni correction to the 21 net-mean p -values, which controls the family-wise error rate at 5%:

- 9 specifications remain positive and significant.

- The **primary** net result does *not* (adjusted $p = 0.052$). Its gross result does (adjusted $p = < 0.0001$).

Holm is conservative here because the 21 tests are highly correlated: the four cost variations share one gross return series. The truth for the net primary test lies between the raw $p = 0.005$ and the adjusted 0.052. We read that as moderate evidence, not decisive evidence.

Harvey, Liu & Zhu (2016) argue that, given the number of factors tested across the literature, a new factor should clear $t \approx 3$. The primary gross t-statistic (4.59) clears that hurdle; the net t-statistic (2.83) does not.

Economic and statistical significance differ. The net Sharpe ratio's bootstrap interval, [0.08, 0.51], runs from negligible to respectable. Statistical evidence that the mean is positive coexists with substantial uncertainty about its size.

IID versus HAC. For the primary series, IID t-statistics are close to the HAC ones (gross: IID 4.45 vs HAC 4.59). The choice matters more for overlapping-cohort variations, where autocorrelation is built in.

9. Subperiods and market states

The three equal-length subperiods were fixed before results:

Period	Months	Gross	Net	Net HAC t	Net 95% CI	Net max DD
1927-07 – 1960-07	397	6.40%	3.29%	0.98	[-3.3%, 9.9%]	-91.0%
1960-08 – 1993-08	397	11.09%	8.32%	3.30	[3.4%, 13.3%]	-36.9%
1993-09 – 2026-08	396	6.34%	3.27%	1.04	[-2.9%, 9.4%]	-60.8%
Pre-July 1963	432	7.21%	4.15%	1.33	[-2.0%, 10.3%]	-91.0%
July 1963 onward	758	8.36%	5.42%	2.58	[1.3%, 9.5%]	-60.8%

The point estimate is positive in every third, gross and net. Only the middle third (1960–1993) is individually significant net of costs. The most recent third, 1993-09 to 2026-08, earned 3.27% net ($t = 1.04$). That is about 33 years of data in which an investor could not have distinguished the net return from zero with confidence.

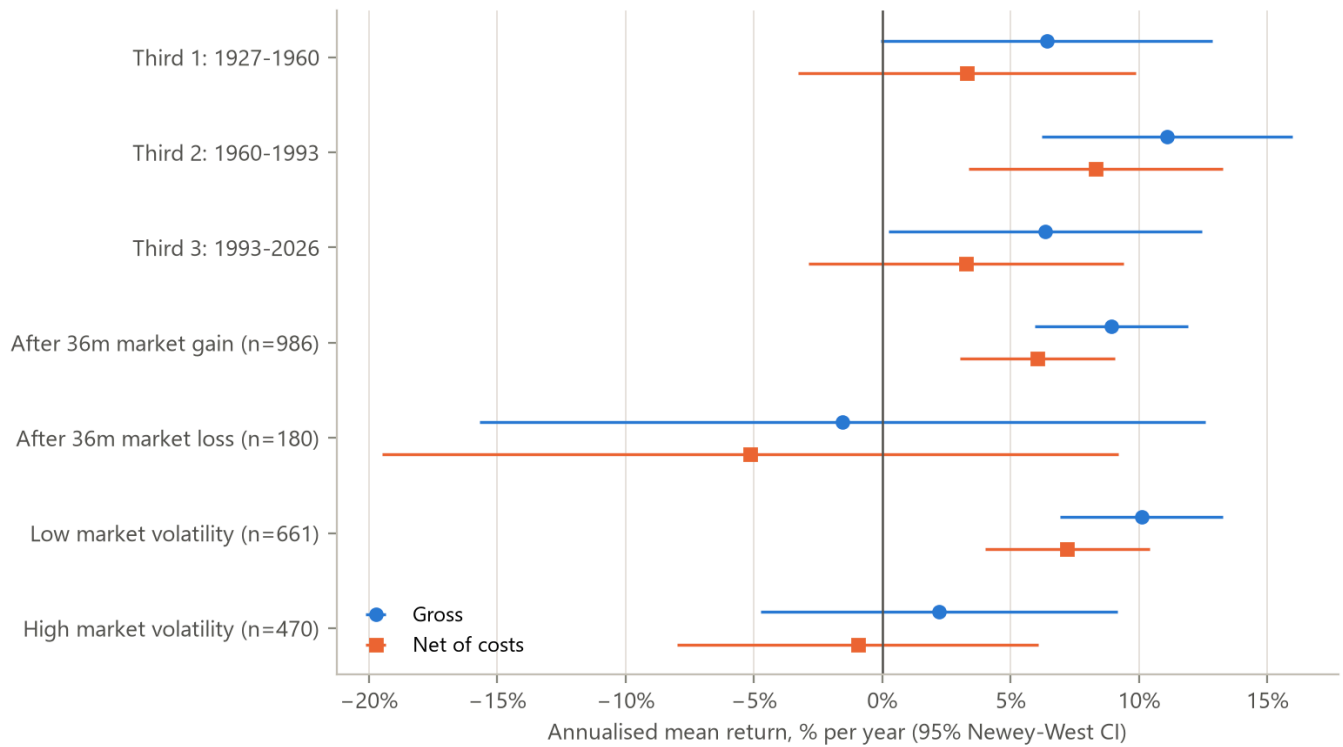
Market states are measured only with information available at formation:

State at formation	Months	Gross	Net (HAC t)
Trailing 36-month market return ≥ 0	986	8.93%	6.04% (3.95)
Trailing 36-month market return < 0	180	-1.54%	-5.15% (-0.70)
Market volatility below its running median	661	10.10%	7.21% (4.42)
Market volatility above its running median	470	2.21%	-0.94% (-0.26)

The whole effect comes from calm or rising markets. After three-year market declines, the point estimate is negative with a very wide interval, consistent with the "momentum crash" pattern described by Cooper, Gutierrez & Hameed (2004) and Daniel & Moskowitz (2016). The DOWN state covers only 180 months, so we treat the conditional estimate as descriptive.

Stability over time and market states

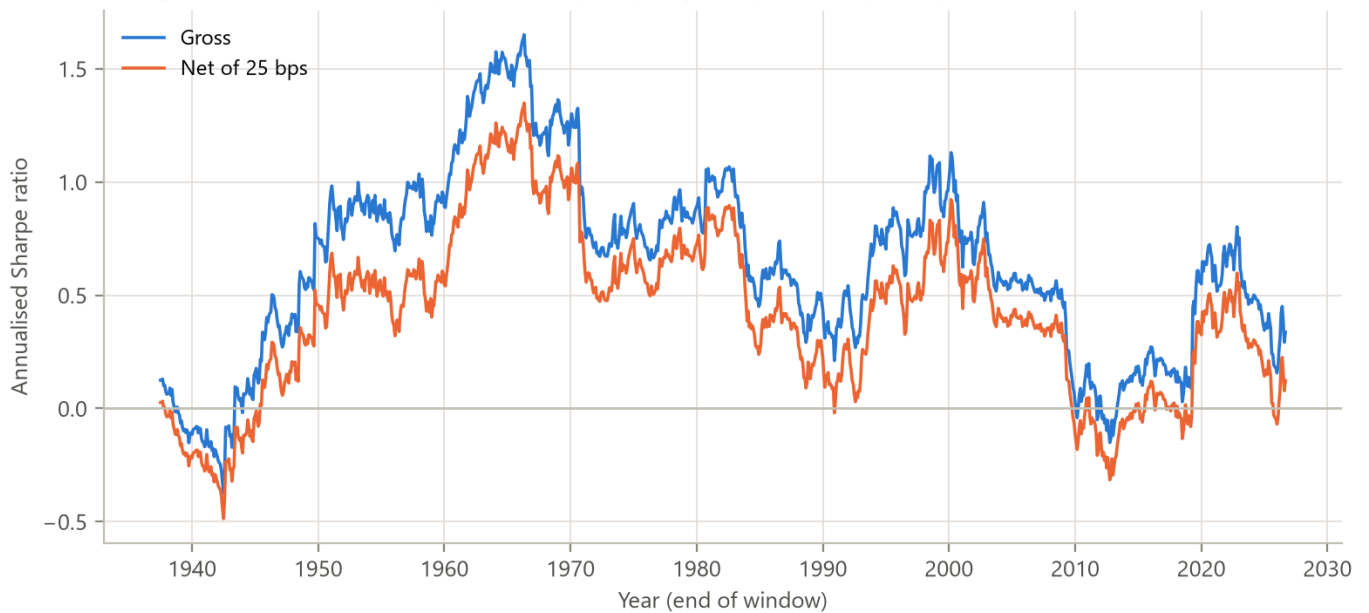
Primary long-short: three equal-length subperiods and states known at formation



Source: Kenneth R. French Data Library (CRSP-based); ScaleX calculations. Sample: holding months 1927-07 to 2026-08. States use only market data through the formation month.

Rolling 10-year Sharpe ratio

Trailing 120-month annualised Sharpe ratio of the primary long-short portfolio (exploratory)



Source: Kenneth R. French Data Library (CRSP-based); ScaleX calculations. Sample: holding months 1927-07 to 2026-08. Overlapping windows: adjacent points share most of their data and are not independent.

10. What does not work

This section is mandatory in every ScaleX publication.

- **Short and medium lookbacks fail after costs.** The 3-1, 3-0, 6-1 and 6-0 signals lose net significance at 25 bps; 3-1 loses money with confidence. A reader who had chosen "6-month momentum", a common convention, would have reached a different conclusion from ours.
- **The primary net result does not survive multiple-testing correction.** Holm-adjusted $p = 0.052$.
- **The pre-stated robustness criterion was not met.** 5 of 16 non-cost variations change the conclusion.
- **Costs above about 67 bps per unit of turnover eliminate the effect** for the primary construction. At 50 bps the net result is no longer significant ($t = 1.11$); at 100 bps it is significantly negative.
- **Crash risk is extreme.** The worst month was -64.4% and the maximum drawdown -91.0%. In the single worst month (1932-08, -63.6% gross), one short position, the 1-firm "Paper" industry (return 300%), alone cost -37.5%. This is a data artefact of thin early industries as much as a property of momentum.
- **The effect vanishes after market declines and in high-volatility states.** This is precisely when it would be most costly to hold.
- **Recent evidence is weak.** The last third of the sample is not individually significant net of costs. The 2000s decade had a negative net mean (exploratory decade table).
- **Inconclusive:** whether the larger net means of deciles or of equal-weighted industry returns are real or an artefact of the flat cost model. Answering that needs a cost model that depends on firm size, which this study does not have.

11. Limitations

- **Industries, not stocks.** These are value-weighted industry baskets, not the individual stocks of the classical momentum literature. Stock-level momentum may behave differently: French's stock-level factor is correlated 0.79 with our primary gross series (exploratory). Stock-level tests need CRSP or equivalent licensed data, which we cannot redistribute. The engine accepts any return panel, so such a test is a configuration change.
- **Survivorship.** Industry portfolios include stocks that later delist, but the treatment of delisting returns is not documented in detail.
- **Thin early industries.** Several pre-1960 "industries" are one or two firms. The exploratory check that excludes them does not change the conclusion.
- **Transaction costs.** The model is a flat, time-invariant charge per unit of turnover. It ignores market impact, bid-ask asymmetry, the trading the industry portfolios do internally, short-sale constraints and borrow fees, financing spreads, and the much higher cost of trading in the early sample. Net figures before about 1980 are almost certainly optimistic.
- **Liquidity and capacity.** We do not model the size that could be traded. Industry portfolios dominated by small stocks, especially under equal weighting, would face meaningful capacity limits.
- **Shorting.** The short leg assumes industries can be shorted at no extra cost throughout 1927–2026. For most of that period, shorting a broad industry basket was impractical or impossible.
- **Model risk.** Results depend on the 4-digit SIC classification, on French's portfolio-construction rules and on our timing conventions. Each is documented, but none is the only reasonable choice.

- **Statistical uncertainty.** The confidence intervals are wide. The 1930s crashes dominate several statistics. HAC inference is asymptotic; with heavy tails (excess kurtosis ≈ 28), finite-sample coverage may be below nominal.
- **External validity.** One country, one asset class, one data source. The sample overlaps heavily with the samples used to discover momentum, so this is not independent confirmation.

12. Conclusion

Does a simple cross-sectional momentum signal remain statistically and economically meaningful under reasonable implementation choices?

Partly. In US industry portfolios from 1927 to 2026, the gross momentum spread is large, monotonic across quintiles and statistically robust. It survives almost every reasonable construction choice we pre-specified.

The economically relevant quantity, the return after trading costs, is much more fragile: - It is moderately significant for the classical 12-1 signal, but not after correcting for the 21 specifications tested. - It disappears for shorter lookbacks. - It is wiped out by costs above roughly 67 bps per unit of turnover. - It has come with severe crashes, concentrated after market declines.

By the criterion we committed to before seeing results, the effect is **partially robust**, not robust. This note does not establish an investable strategy, and should not be read as one.

Reproducibility

Everything in this note can be regenerated from public inputs:

```
git clone https://github.com/scalex-research/scalex-momentum-research
cd scalex-momentum-research
python -m venv .venv && source .venv/bin/activate # Windows: .venv\Scripts\activate
pip install -r requirements-lock.txt && pip install -e . --no-deps
python scripts/download_data.py # fetches the French Data Library files (not redistributed)
python scripts/run_experiment.py # all tables + results/summary.json
python scripts/generate_figures.py # all figures
python scripts/build_paper.py # this note, with every number filled from summary.json
pytest # full test suite incl. look-ahead canaries
```

French revises the library as CRSP data are updated. A later download can therefore differ slightly from the 202608 vintage used here; compare your `data/raw/MANIFEST.json` hashes with `results/run_metadata.json`. See [docs/reproducibility.md](https://github.com/scalex-research/scalex-momentum-research/blob/main/docs/reproducibility.md).

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<https://github.com/scalex-research/scalex-momentum-research>